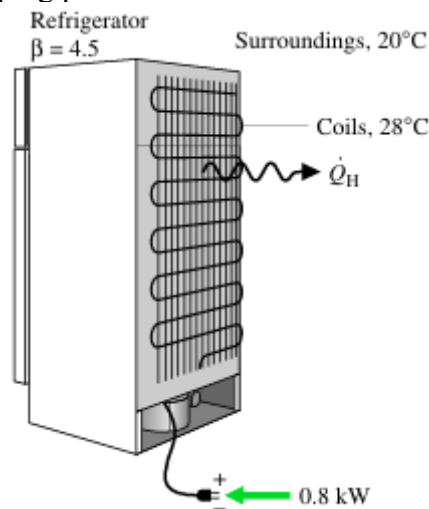


Second Law of Thermodynamics

- (1) The refrigerator shown in Fig. operates at steady state with a coefficient of performance of 4.5 and a power input of 0.8 kW. Energy is rejected from the refrigerator to the surroundings at 20°C by heat transfer from metal coils whose average surface temperature is 28°C. Determine:
- the rate energy is rejected, in kW.
 - the lowest theoretical temperature inside the refrigerator, in K.
 - the maximum theoretical power, in kW, that could be developed by a power cycle operating between the coils and the surroundings. Would you recommend making use of this opportunity for developing power?



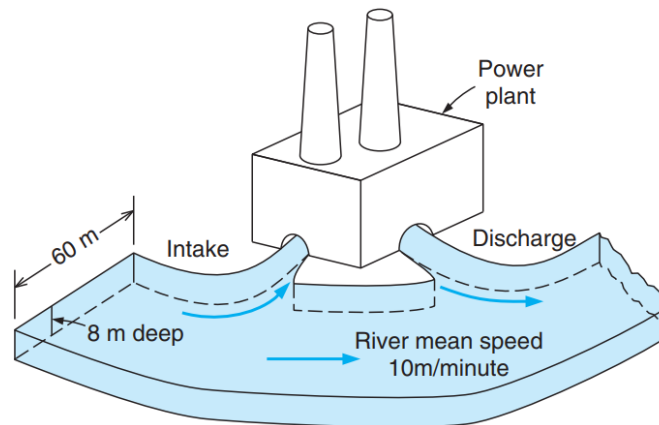
- (2) The preliminary design of a space station calls for a power cycle that at steady state receives energy by heat transfer at $T_H = 600$ K from a nuclear source and rejects energy to space by thermal radiation. For the radiative surface, the temperature is T_C , the emissivity is 0.6, and the surface receives no radiation from any source. The thermal efficiency of the power cycle is one-half that of a reversible power cycle operating between reservoirs at T_H and T_C .
- For $T_C = 400$ K, determine the net power developed per unit of radiator surface area, in kW/m², and the thermal efficiency.
 - Plot $\dot{W}_{cycle}/A$ and thermal efficiency versus T_C , and determine the maximum value of $\dot{W}_{cycle}/A$.
 - Determine the range of temperatures T_C , in K, for which $\dot{W}_{cycle}/A$ is within 2 percent of the maximum value obtained in part (b).

The Stefan–Boltzmann constant is $5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$

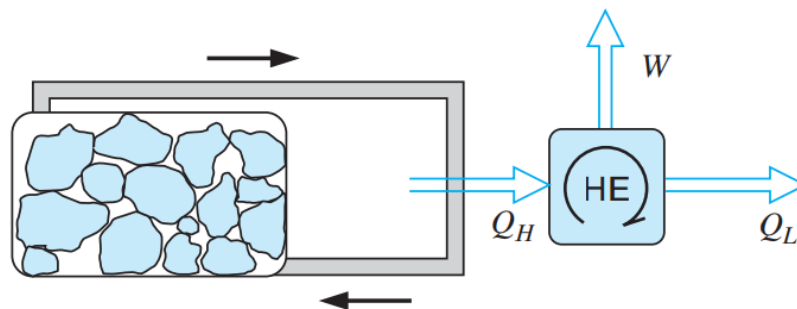
- (3) One kilogram of air as an ideal gas executes a Carnot power cycle having a thermal efficiency of 60%. The heat transfer to the air during the isothermal expansion is 40 kJ. At the end of the isothermal expansion, the pressure is 5.6 bar and the volume is 0.3 m³. Determine:
- the maximum and minimum temperatures for the cycle, in K.
 - the pressure and volume at the beginning of the isothermal expansion in bar and m³, respectively.
 - the work and heat transfer for each of the four processes, in kJ.

(d) Sketch the cycle on a $p - v$ coordinates.

- (4) A car engine operates with a thermal efficiency of 35%. Assume the air conditioner has a COP of $\beta = 3$ working as a refrigerator cooling the inside using engine shaft work to drive it. How much extra fuel energy should be spent to remove 1 kJ from the inside?
- (5) A proposal is to build a 1000-MW electric power plant with steam as the working fluid. The condensers are to be cooled with river water (see Figure below). The maximum steam temperature is 550°C, and the pressure in the condensers will be 10 kPa. Estimate the temperature rise of the river downstream from the power plant.

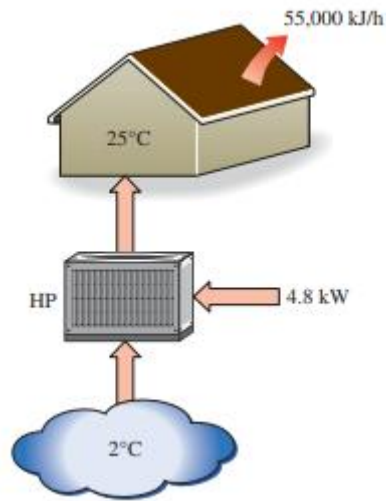


- (6) Thermal storage is made with a rock (granite) bed of 2 m³ that is heated to 400 K using solar energy. A heat engine receives a Q_H from the bed and rejects heat to the ambient at 290 K. The rock bed therefore cools down, and as it reaches 290 K the process stops. Find the energy the rock bed can give out. What is the heat engine efficiency at the beginning of the process and what is it at the end of the process?

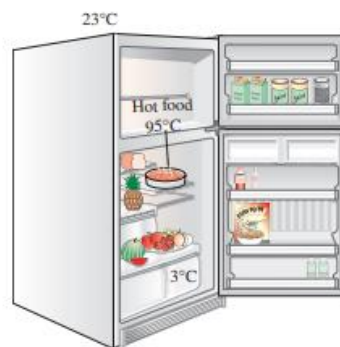


- (7) A coal-burning steam power plant produces a net power of 300 MW with an overall thermal efficiency of 32 percent. The actual gravimetric air-fuel ratio in the furnace is calculated to be 12 kg air/kg fuel. The heating value of the coal is 28,000 kJ/kg. Determine (a) the amount of coal consumed during a 24-hour period and (b) the rate of air flowing through the furnace.
- (8) A Carnot heat pump is to be used to heat a house and maintain it at 25°C in winter. On a day when the average outdoor temperature remains at about 2°C, the house is estimated to lose heat at a rate of 55,000 kJ/h. If the heat pump consumes 4.8 kW of power while operating, determine (a) how long the heat pump ran on that day; (b) the total heating

costs, assuming an average price of \$0.11/kWh for electricity; and (c) the heating cost for the same day if resistance heating is used instead of a heat pump.



- (9) It is commonly recommended that hot foods be cooled first to room temperature by simply waiting a while before they are put into the refrigerator to save energy. Despite this commonsense recommendation, a person keeps cooking a large pan of stew three times a week and putting the pan into the refrigerator while it is still hot, thinking that the money saved is probably too little. But he says he can be convinced if you can show that the money saved is significant. The average mass of the pan and its contents is 5 kg. The average temperature of the kitchen is 23°C, and the average temperature of the food is 95°C when it is taken off the stove. The refrigerated space is maintained at 3°C, and the average specific heat of the food and the pan can be taken to be 3.9 kJ/kg·°C. If the refrigerator has a coefficient of performance of 1.5 and the cost of electricity is \$0.125/kWh, determine how much this person will save a year by waiting for the food to cool to room temperature before putting it into the refrigerator.



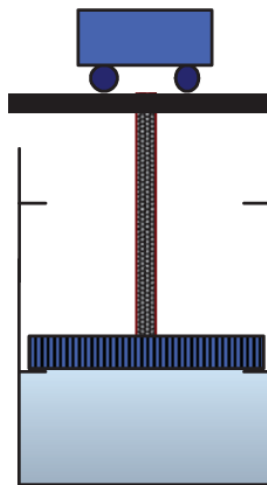
- (10) An inventor claims to have a device that uses a solar water heater for air conditioning, without the need for work input. The inventor states essentially that given a supply of hot water at 95°C and an environment at 38°C the device can remove heat at a rate of 3.4 kW from a room at 20°C.
- (a) Is such a device feasible?
- (b) Assuming it is feasible and ideal, find the rate of heat supply to the water heater.
- (11) A solar collector is used as a heat source for a Carnot engine with a heat sink at 300 K. The efficiency of the solar collector ϵ is defined as the fraction of the energy reaching the

collector that is actually absorbed. It is related to the temperature of the collector as follows:

$$\varepsilon = 0.75 - 1.75(T/300 - 1)$$

Determine the best operating temperature of the collector, that is, the temperature that yields maximum power from the Carnot engine.

- (12) A cylinder shown in Figure contains 0.5 kg of H₂O at 40°C and 150 kPa. It is covered with a piston of a weight equivalent to a pressure of 150 kPa. At the beginning of the process the piston rests on the lower stop. A weight equivalent to 450 kPa is added to the piston, and then the cylinder is heated by a heat reservoir at 250°C raising the pressure of the system till its temperature reaches 200°C. At this point the piston reaches the upper stop. The extra weight is then removed and the cylinder is brought in contact with a cold reservoir at 10°C. The heat removal is stopped when the temperature in the system reaches 40°C, thus completing a cycle. (a) Find the heat and work interactions of the cycle. (b) Find the efficiency of the cycle.



- (13) When one junction of a thermoelectric generator is in good contact with the environment at 25°C and the other with a reservoir at 300°C, the open circuit voltage is 0.95 V. An ideal 50 W electric motor connected to the generator draws 80 A. The rate of heat removal from the cold junction is 900 W.
- Find the internal resistance of the generator.
 - Find the efficiency of the generator.
 - Is this cycle reversible?
- (14) A p–v diagram of an imaginary heat engine is represented in the figure by a circle of 16 cm in diameter. Each centimeter of the pressure axis corresponds to 100 kPa and 0.1 m³/kg for the specific volume axis.
- Find the work performed per cycle by 1 kg of working fluid.
 - Find the engine efficiency if it rejects 4000 kJ/kg during each cycle.

